

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 09 · Momentum, Impulse and Collisions

Questions

Student book · questions with writing space

- 28 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 09

Momentum, Impulse & Collisions

Your classified route

16

QUESTIONS

101

MARKS

2

FAMILIES

7

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 09 | FAMILY 01

Conservation of momentum in a direct collision

WHAT THIS FAMILY ASKS

Apply signed one-dimensional momentum conservation to two particles.

VARIANT ROUTE

V01 Particles separate after collision

V02 Particles coalesce

V03 Unknown mass/speed parameter or ratio

V04 Unknown post-collision direction

8 classified questions

57 total marks

1. Two particles, P and Q , of mass m_1 and m_2 respectively, are moving on a smooth horizontal plane. The particles are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision, both particles are moving with speed u .

The direction of motion of each particle is reversed by the collision.

Immediately after the collision, the speed of Q is $\frac{1}{3}u$.

- (a) Find, in terms of m_2 and u , the magnitude of the impulse exerted by P on Q in the collision.

(3)

DR ESAM AHMED

Two particles, P and Q , of mass m_1 and m_2 respectively, are moving on a smooth horizontal plane. The particles are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision, both particles are moving with speed u .

The direction of motion of each particle is reversed by the collision.

Immediately after the collision, the speed of Q is $\frac{1}{3}u$.

(b) Find, in terms of m_1 , m_2 and u , the speed of P immediately after the collision. (3)

(c) Hence show that $m_2 > \frac{3}{4}m_1$ (2)

DR ESAM AHMED

2. Two particles, A and B , are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle A has mass $3m$ kg and particle B has mass m kg.

Immediately before the collision, both particles have a speed of 1.5 m s^{-1}

Immediately after the collision, the direction of motion of A is unchanged and the difference between the speed of A and speed of B is 1 m s^{-1}

- (a) Find
- (i) the speed of A immediately after the collision,
 - (ii) the speed of B immediately after the collision.

(5)

DR ESLAM AHMED

Two particles, A and B , are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle A has mass $3m$ kg and particle B has mass m kg.

Immediately before the collision, both particles have a speed of 1.5 m s^{-1}

Immediately after the collision, the direction of motion of A is unchanged and the difference between the speed of A and speed of B is 1 m s^{-1}

(b) Find, in terms of m , the magnitude of the impulse exerted on B in the collision.

(3)

DR ESAM AHMED

1. A particle A has mass 4 kg and a particle B has mass 2 kg .

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\text{ ms}^{-1}$ and the speed of B is $3u\text{ ms}^{-1}$

Immediately after the collision, the speed of B is $2u\text{ ms}^{-1}$

The direction of motion of B is reversed by the collision.

- (a) Find, in terms of u , the speed of A immediately after the collision.

(3)

- (b) State the direction of motion of A immediately after the collision.

(1)

DR ESAM AHMED

A particle A has mass 4 kg and a particle B has mass 2 kg .

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\text{ ms}^{-1}$ and the speed of B is $3u\text{ ms}^{-1}$

Immediately after the collision, the speed of B is $2u\text{ ms}^{-1}$

The direction of motion of B is reversed by the collision.

- (c) Find, in terms of u , the magnitude of the impulse received by B in the collision.
State the units of your answer.

(3)

DR ESAM AHMED

1. Particle A has mass $4m$ and particle B has mass $3m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Immediately **before** the collision, the speed of A is $2x$ and the speed of B is x .

Immediately **after** the collision, the speed of A is y and the speed of B is $5y$.

The direction of motion of each particle is reversed as a result of the collision.

- (a) Show that $y = \frac{5}{11}x$.

(3)

DR ESLAM AHMED

Particle A has mass $4m$ and particle B has mass $3m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Immediately **before** the collision, the speed of A is $2x$ and the speed of B is x .

Immediately **after** the collision, the speed of A is y and the speed of B is $5y$.

The direction of motion of each particle is reversed as a result of the collision.

- (b) Find, in terms of m and x , the magnitude of the impulse received by A in the collision.

(3)

DR ESAM AHMED

1. Railway truck A has mass $5m$ and railway truck B has mass $3m$. The trucks are moving towards each other in opposite directions along the same straight track when they collide directly. Immediately before the collision the speed of A is u and the speed of B is $1.5u$

As a result of the collision, the trucks couple together.

- (a) Find, in terms of u , the common speed of the trucks immediately after the collision. (3)
- (b) Find, in terms of m and u , the magnitude of the impulse exerted on B by A in the collision. (3)

DR ESAM AHMED

2. A particle P has mass km and a particle Q has mass m . The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P has speed $3u$ and Q has speed u .

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

- (a) Find the value of k .

(4)

DR ESAM AHMED

A particle P has mass km and a particle Q has mass m . The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P has speed $3u$ and Q has speed u .

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on Q in the collision. (3)

DR ESAM AHMED

1. A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

- (a) Find, in terms of m and u , the magnitude of the impulse exerted on Q by P in the collision.

(2)

DR ESAM AHMED

A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

(b) Find the two possible values of k .

(5)

DR ESAM AHMED

2.

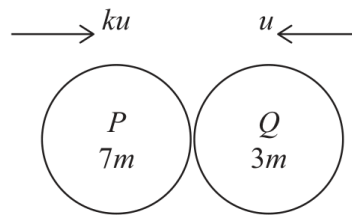


Figure 2

Particle P of mass $7m$ and particle Q of mass $3m$ are moving in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is ku and the speed of Q is u , as shown in Figure 2.

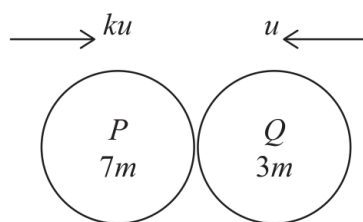
Immediately **after** the collision, the speed of P is w and the speed of Q is $2w$.

The direction of motion of Q is reversed by the collision.

The impulse received by Q in the collision has magnitude $\frac{7}{2}mu$.

(a) Find w in terms of u .

(3)

**Figure 2**

Particle P of mass $7m$ and particle Q of mass $3m$ are moving in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is ku and the speed of Q is u , as shown in Figure 2.

Immediately **after** the collision, the speed of P is w and the speed of Q is $2w$.

The direction of motion of Q is reversed by the collision.

The impulse received by Q in the collision has magnitude $\frac{7}{2}mu$.

(b) Find the two possible values of k .

(5)

TOPIC 09 | FAMILY 02

Impulse as change in momentum

WHAT THIS FAMILY ASKS

Use $I = m(v - u)$ with a declared positive direction and distinguish vector/sign from magnitude.

VARIANT ROUTE

V01 Impulse on one particle

V02 Given impulse to recover velocity

V03 Slack string becomes taut impulsively

8 classified questions

44 total marks

1. A railway truck S of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck T of mass 30 tonnes which is at rest. Immediately before the collision the speed of S is 4 m s^{-1}

As a result of the collision, the two railway trucks join together.

Find

- (a) the common speed of the railway trucks immediately after the collision,

(2)

- (b) the magnitude of the impulse exerted on S in the collision, stating the units of your answer.

(3)

DR ESAM AHMED

2.

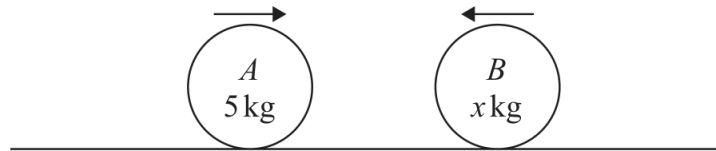
**Figure 2**

Figure 2 shows two particles, A and B , moving in opposite directions on a smooth horizontal surface. Particle A has mass 5 kg and particle B has mass $x\text{ kg}$.

The particles collide directly.

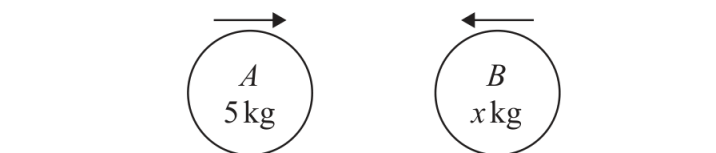
Immediately before the collision, the speed of A is 3 ms^{-1} and the speed of B is $x\text{ ms}^{-1}$

Immediately after the collision, the speed of A is 1 ms^{-1} and its direction of motion is unchanged.

Immediately after the collision, the speed of B is 1.5 ms^{-1}

(a) Find the value of x .

(3)

**Figure 2**

The particles collide directly.

Immediately before the collision, the speed of A is 3 m s^{-1} and the speed of B is $x \text{ m s}^{-1}$

Immediately after the collision, the speed of A is 1 m s^{-1} and its direction of motion is unchanged.

Immediately after the collision, the speed of B is 1.5 m s^{-1}

(b) Find the magnitude of the impulse exerted on A by B in the collision.

(3)

DR ESAM AHMED

2. Two particles, P and Q , have masses $2m$ and m respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $2u$.

The magnitude of the impulse exerted on Q by P in the collision is $5mu$.

Find

- (a) the speed of P immediately after the collision, (3)

- (b) the speed of Q immediately after the collision. (3)

DR ESAM AHMED

1. Two particles, P and Q , are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of Q is $2u$. The mass of Q is $3m$ and the magnitude of the impulse exerted by P on Q in the collision is $4mu$.

Find

- (a) the speed of Q immediately after the collision,

(3)

- (b) the direction of motion of Q immediately after the collision.

(1)

DR ESAM AHMED

1.

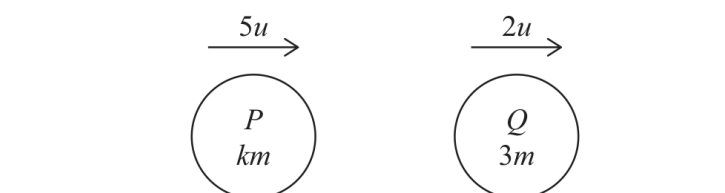
**Figure 1**

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(a) Find v in terms of u only.

(3)

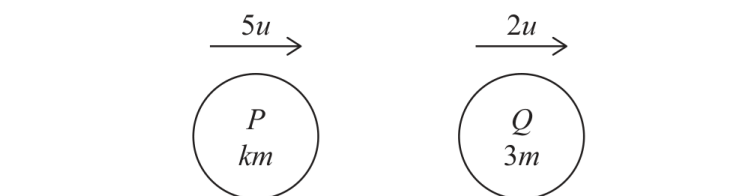
**Figure 1**

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(b) Find the value of k

(3)

1. A particle P of mass $3m$ and a particle Q of mass $5m$ are on a smooth horizontal surface. The particles move towards each other in opposite directions along the same straight line and collide.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $4u$

The magnitude of the impulse exerted on P by Q in the collision is $\frac{33}{2}mu$

Find

- (i) the speed of P immediately after the collision,
- (ii) the speed of Q immediately after the collision.

(6)

DR ESAM AHMED

1. Two particles, P and Q , have masses $3m$ and $2m$ respectively. The particles are connected by a light inextensible string. Initially P and Q are at rest on a smooth horizontal plane with the string slack.

Particle P is then projected along the plane directly away from Q with speed $4u$. At the same instant, particle Q is projected along the plane in the opposite direction with speed $3u$.

Find

- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

DR ESAM AHMED

1. Two particles, A and B , have masses m and $3m$ respectively. The particles are connected by a light inextensible string. Initially A and B are at rest on a smooth horizontal plane with the string slack.

Particle A is then projected along the plane away from B with speed U .

Given that the common speed of the particles immediately after the string becomes taut is S

- (a) find S in terms of U .

(2)

- (b) Find, in terms of m and U , the magnitude of the impulse exerted on A immediately after the string becomes taut.

(3)

DR ESAM AHMED